Geography of Tamil Nadu and its impact on economic growth

Table of Contents

- Formation of Tamil Nadu
- Location and Size
- Boundaries and Neighbours
- Physiographic Divisions of Tamil Nadu (Main Topic)
- Major Physiographic Divisions of Tamil Nadu
  - Western Ghats
  - Eastern Ghats
  - Plateaus
  - Inland Plains
  - Coastal Plains
- Climate of Tamil Nadu
- Soils of Tamil Nadu
- Types of Forests in Tamil Nadu
- Wildlife of Tamil Nadu

Introduction

- ✓ Understand own region → global perspective
- ✓ Rich cultural heritage
- ✓ Unique physiography
- ✓ Features: beaches, waterfalls, hills, forests, flora & fauna

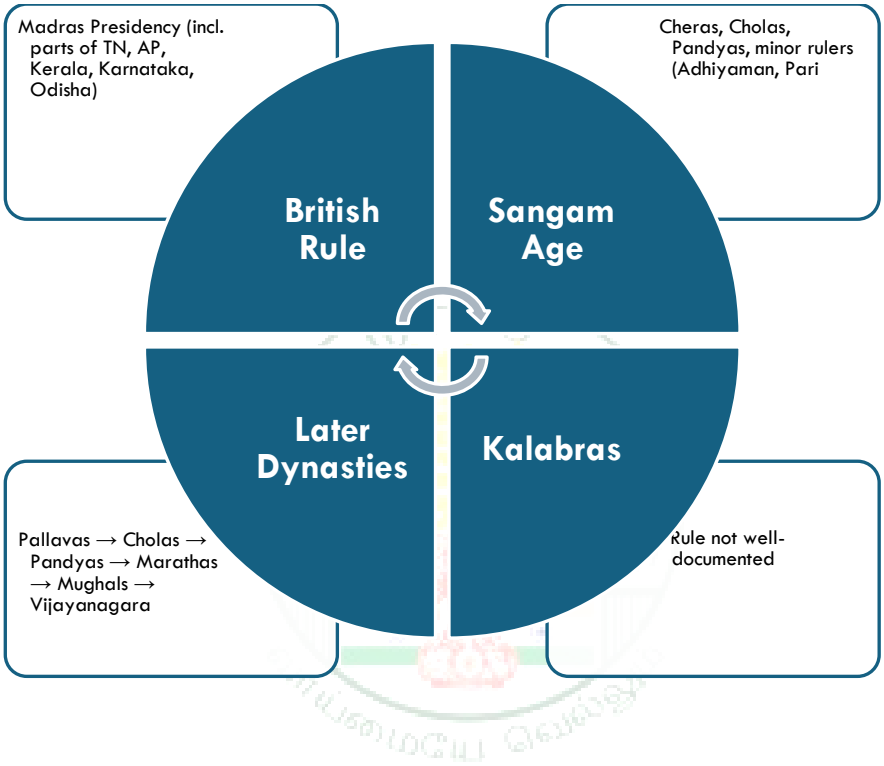
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Formation of Tamil Nadu



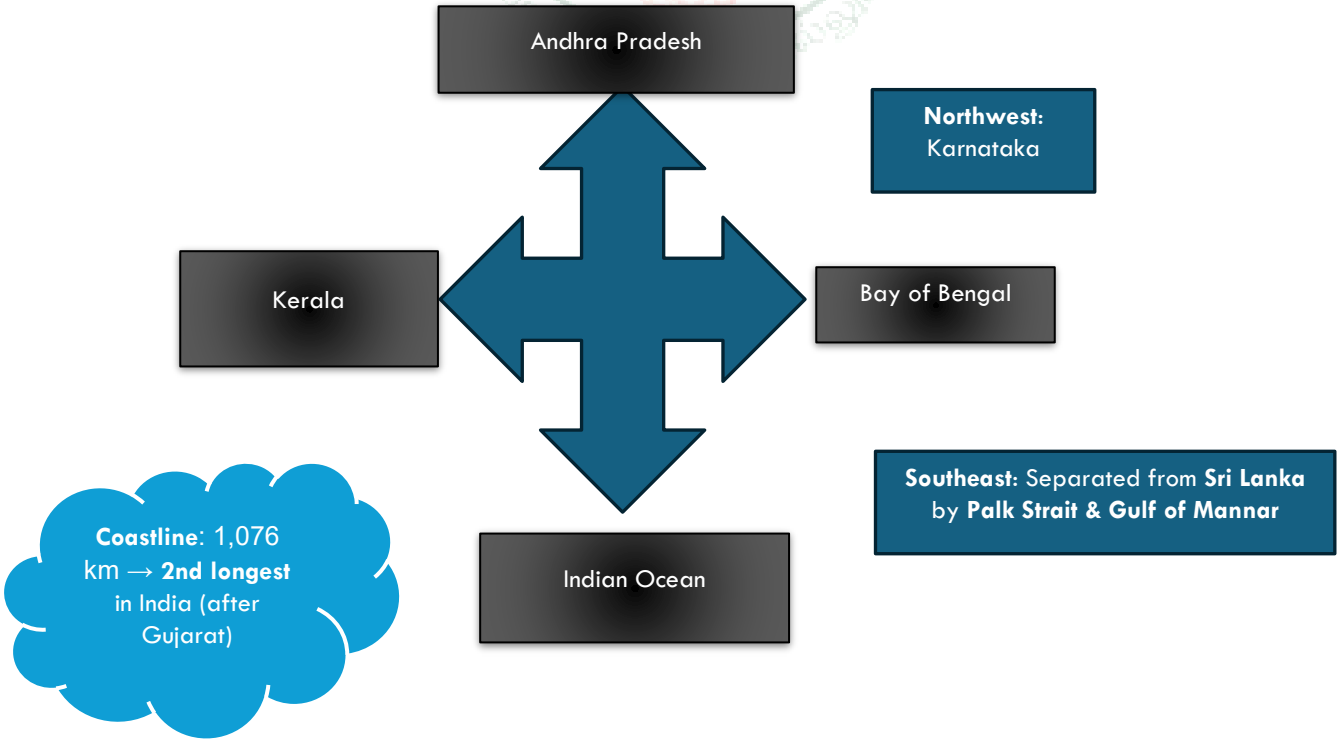
Post-Independence:

- Linguistic reorganisation
- Telugu areas separated
- 13 districts initially
- Renamed *Tamil Nadu* on **14 Jan 1969** by **C.N. Annadurai**

Location and Size

Latitudes: 8°4'N to 13°35'N
Longitudes: 76°18'E to 80°20'E
Extent:
East: Point Calimere
West: Anaimalai Hills
North: Pulicat Lake
South: Cape Comorin (Kanyakumari)
Area: 1,30,058 sq.km
11th largest state
Covers **4%** of India's area

Boundaries and Neighbours



Physiographic Divisions of Tamil Nadu (Main Topic)

- ✓ Part of **Deccan Plateau**
- ✓ Ancient **Gondwana Land**
- ✓ Features: **eroded mountains, valleys, plains**
- ✓ Slope: **west → east**
- ❖ Major Divisions:
  - **Western Ghats**
  - **Eastern Ghats**
  - **Plateau**
  - **Coastal Plains**
  - **Inland Plains**

1. Western Ghats

- **Location:** Nilgiris (N) to Marunthuvazh Malai (S)
- **Height:** 2,000–3,000 m | **Area**: 2,500 sq.km
- **Major Passes**: Palghat, Shencottah, Aralvaimozhi, Achankoil

Important Hills:

Nilgiris	■Peaks: Doddabetta (2,637 m), Mukkuruthi (2,554 m) ■Hill stations: Ooty, Coonoor ■Flora: 2,700+ flowering plants Fauna: **Nilgiri Tahr**
Anaimalai	Border of TN–Kerala Anaimalai Tiger Reserve, Valparai, Kadamparai HEP
Palani Hills	District: Dindigul Peaks: Vandaravu (2,533 m), Kodaikanal (2,150 m)
Cardamom Hills (Yela Malai)	Crops: Cardamom, Pepper, Coffee
Varusanadu & Andipatti Hills	Waterfalls: Suruli, Kumbakarai Sanctuary: Srivilliputhur Grizzled Squirrel Rivers: Vaigai & tributaries
Pothigai Hills	Names: Agasthiyar Malai, Southern Kailash Forests, temples, biodiversity Reserve: Kalakkad Mundanthurai TR
Mahendragiri Hills	• Border: Kanyakumari & Tirunelveli Avg. height: 1,645 m Facility: ISRO Propulsion Complex

🌰 2. Eastern Ghats

- **Nature**: Discontinuous & Irregular
- **Height:** 1,100–1,600 m
- **Rivers dissect through to Bay of Bengal**

Major Hills

Javadhu Hills:
Districts: Vellore, Tiruvannamalai
Peak: Melpattu (~1,150 m)
Observatory: Vainu Bappu (VBO)
Known for: Sandalwood, herbs

Kalvarayan Hills
Name from 'Karalar'
•District: Kallakurichi
•Sections: Chinna (825 m), Periya (1,220 m

Servarayan Hills
District: Salem
•Peak: Solaikaradu (1,620 m)
•Hill Station: Yercaud ("Poor man's Ooty")
•Temple: Servarayan Temple (1,623 m)

Kolli Hills
District: Namakkal
Area: 2,800 sq.km | Height: 1,300 m
Forests: Shola cover
Agriculture: Coffee, fruits, flowers

Servarayan Hills
Districts: Perambalur, Salem, Tiruchirappalli
Name: "Green Hills"
Product: Jackfruit

3. District-wise Hills

District	Major Hills
Coimbatore	Maruthamalai, Velliangiri, Anaimalai
Dindigul	Palani, Kodaikanal
The Nilgiris	Nilgiris Hills
Vellore	Javadhu, Yelagiri, Rathinamalai
Namakkal	Kolli Hills
Salem	Servarayan, Kanjamalai, Chalk Hills

Kallakurichi	Kalvarayan
Villupuram	Gingee Hills
Tirunelveli	Agasthiyarmalai, Mahendragiri
Kanyakumari	Marunthuvazh Malai
Dharmapuri	Theerthamalai, Chitteri, Vathalmalai
Erode	Chenni Hills, Sivan Hills



Major Physiographic Divisions of Tamil Nadu

Division	Key Features
Western Ghats	High eroded mountains (2,000–3,000 m), continuous range, passes like Palghat, Shencottah

Eastern Ghats	Discontinuous hills (1,100–1,600 m), dissected by rivers
Plateaus	Between Ghats; triangular; 150–600 m elevation
Inland Plains	Riverine plains of Palar, Ponnaiyar, Cauvery, Thamirabarani
Coastal Plains	Coromandel/Cholamandalam coast (Chennai–Kanyakumari), includes Teri dunes & coral rocks

2. Western Ghats – Major Hills & Features

Hills/Region	Location	Highlights
Nilgiri Hills	NW TN	Doddabetta (2,637 m), Mukkuruthi (2,554 m), Ooty & Coonoor
Anaimalai Hills	Border of TN-Kerala	Valparai, Tiger Reserve, Aliyar Dam
Palani Hills	Dindigul	Vandaravu (2,533 m), Kodaikanal (2,150 m)
Cardamom Hills	SW TN	Grows cardamom, coffee, pepper
Varusanadu & Andipatti Hills	Near Madurai	Megamalai, Kurangani, Suruli Falls
Pothigai Hills	Tirunelveli	Rich biodiversity, Agasthiyar legend
Mahendragiri Hills	Tirunelveli-Kanyakumari	ISRO propulsion complex, 1,645 m

3. Eastern Ghats – Major Hills

Hill Range	Districts	Special Features

Javadhu Hills	Vellore, Tiruvannamalai	Melpattu peak; Sandalwood decline
Kalvarayan Hills	Kallakurichi	Periya (1,220 m) & Chinna (825 m) Kalvarayan
Servarayan Hills	Salem	Yercaud, Shervarayan Temple (1,623 m)
Kolli Hills	Namakkal	Arpaleeswarar Temple, Shola forest
Pachaimalai	Perambalur, Salem, Trichy	Green vegetation, jackfruit

4. Plateau Regions

Plateau	Districts	Elevation & Features
Bharamahal Plateau	Dharmapuri, Krishnagiri	Part of Mysore Plateau, 350–710 m
Coimbatore Plateau	Salem, Coimbatore, Erode	Bhavani, Noyyal, Amaravathi rivers
Sigur Plateau	Nilgiris	Intermontane plateau
Madurai Plateau	Madurai	Vaigai & Thamirabarani basins

5. Plains of Tamil Nadu

Type	Key Features
Inland Plains	Drained by Palar, Ponnaiyar, Cauvery, Thamirabarani
Cauvery Plains	Very fertile – Salem to Nagapattinam
Coastal Plains (Coromandel)	Chennai to Kanyakumari; includes Teri dunes & coral reefs in Gulf of Mannar

6. Famous Beaches

Marina Beach (Chennai)
Elliot's Beach (Chennai)
Kovalam Beach (Kanyakumari)
Silver Beach (Cuddalore)

7. River Drainage of Tamil Nadu

River	Origin	Major Features
Cauvery	Talacauvery (Karnataka)	Mettur Dam, Kallanai, Delta – "Garden of Southern India"
Palar	Kolar (Karnataka)	Flows through Vellore, Kancheepuram
Ponnaiyar (Then Pennaiyar)	Nandi Hills (Karnataka)	Joins Bay of Bengal near Puducherry
Vaigai	Varusanadu Hills	Drains Madurai, Sivagangai, Ramanathapuram
Thamirabarani	Pothigai Hills	Only perennial river; joins Bay of Bengal at Thoothukudi

Climate of Tamil Nad

Location	Lies **south of the Tropic of Cancer** (near Equator)
Climate Type	• Predominantly **Tropical Maritime** (East coast); **Mountain Climate** in hills
Annual Temperature	**18°C to 43°C**
Annual Rainfall	**958.5 mm**
Influences	**Bay of Bengal, Indian Ocean, Western Ghats, Altitude**
Special Notes	Interior regions are hot & dry; hills are cool & pleasant

Seasons in Tamil Nadu

Season	Months	Temperature / Rainfall	Special Features
Winter	Jan – Feb	15°C to 25°C; Hill stations: Below 5°C, even 0°C	Slanting sunrays; dry & cool; mist and frost in hills
Summer	Mar – May	30°C to >40°C	Vertical sunrays; Mango/Blossom showers in May
Southwest Monsoon	Jun – Sep	50–100 cm (West TN)	Rain shadow effect; less rainfall in eastern TN

Northeast Monsoon *(Main Rainy Season)*	Oct – mid-Dec	Coastal: 100–200 cm Inland: 50–100 cm	48% of annual rainfall; cyclones frequent; major for agriculture

Monsoons of Tamil Nadu

Monsoon	Direction	Rainfall	Characteristics
Southwest Monsoon	SW (from Arabian Sea)	Low (esp. east TN)	Rain shadow region; only western/southern districts like Nilgiris, Kanyakumari get moderate rain
Northeast Monsoon	NE (from Bay of Bengal)	High – esp. east coast	Main rainy season; caused by high pressure over Central Asia; influenced by Coriolis force

Cyclones in Tamil Nadu



Special Climatic Zones

Region	Climate Type	Notable Places
East Coast	Tropical Maritime	Chennai, Cuddalore
Interior	Hot & Dry	Trichy, Madurai
Hills	Mountain Climate	Nilgiris, Kodaikanal, Anaimalai
Wettest Place	Chinnakallar (near Valparai)	3rd wettest in India, wettest in TN

Important Concepts

Term	Meaning
Coriolis Force	Deflects winds to the right in Northern Hemisphere due to Earth's rotation
Retreating Monsoon	Another name for Northeast Monsoon
Rain Shadow Region	Area (like TN) receiving less rainfall due to obstruction of wind by Western Ghats

Soils of Tamil Nadu

Type of Soil	Formation / Nature	Key Features	Crops Grown	Major Districts

Alluvial Soil	Formed by river silt deposits	Porous, loamy, contains lime, potassium, magnesium but low in nitrogen and humus	Paddy, sugarcane, banana, turmeric	Thanjavur, Tiruvarur, Nagapattinam, Cuddalore, Villupuram, Tirunelveli, Kanyakumari
Black Soil *(Regur)*	Formed by weathering of igneous rocks, Deccan lava	Clayey, fine-textured, rich in lime, potash, but poor in nitrogen & phosphoric acid	Cotton, sorghum, cumbu, fodder crops	Coimbatore, Madurai, Virudhunagar, Tirunelveli, Thoothukudi
Red Soil	Weathered crystalline rocks; oxidation of iron	Sandy-loamy, porous, non-retentive, poor in nutrients	Ragi, tobacco, vegetables, paddy (with manure)	Sivagangai, Ramanathapuram, and widespread in central TN
Laterite Soil	Formed by intense leaching in heavy rainfall zones	Rich in iron/aluminium oxides; acidic	Tea, coffee, pepper, ginger, plantain, paddy	Patches in Kancheepuram, Tiruvallur, Thanjavur, Nilgiris
Saline Soil	Due to salt accumulation, worsened by 2004 tsunami	Saline, infertile, not suitable for cultivation	Very low cultivation	Vedaranyam, Coromandel coast (post-tsunami affected)

Soil Erosion in Tamil Nadu

Factor	Impact
Causes	Deforestation, overgrazing, urbanisation, heavy rain
Effects	Loss of topsoil and fertility → reduced agricultural productivity

Desertification	Affects ~12% of Tamil Nadu's land (as per ISRO)
Worst-Affected Districts	Theni, Nilgiris, Kanyakumari
Sand Deposition	~12,000 hectares in Theni and Rajapalayam due to wind and runoff

Types of Forests in Tamil Nadu

Forest Type	Features	Main Districts	Major Species
Tropical Evergreen Forest	Dense, multi-layered, found in high rainfall areas	Western Ghats: Tirunelveli, Kanyakumari, Nilgiris, Coimbatore	Cinnamon, Jack, Jamun, Ironwood, Kindal
Semi-Evergreen Forest	Found in sub-tropical climates	Eastern Ghats: Servarayan, Kollimalai, Pachaimalai	Indian Mahogany, Mango, Jack, Monkey Teak
Montane Temperate Forest (Sholas)	Found at altitudes above 1000 m, evergreen short trees	Nilgiris, Anaimalai, Palani Hills	Rose Apple, Nilgiri Champa, Wight's Litsea
Tropical Deciduous Forest	Trees shed leaves in dry season, medium height	Margins of evergreen forests	Silk Cotton, Kadamba, Dog Teak, Bamboo
Mangrove Forest	Found in coastal and deltaic regions, salt-tolerant, protect coast	Pichavaram (Cuddalore), Vedaranyam, Muthupet, Thoothukudi	Avicennia, Rhizophora, Wild Jasmine

Tropical Thorn Forest	Grows in dry regions with less than 400 m altitude	Dharmapuri, Ramanathapuram, Virudhunagar	Neem, Acacia, Palm, Shrubs

Mangroves

Role - Prevents coastal erosion, protects coral reefs, supports biodiversity

Pichavaram Mangroves - Near Chidambaram (Cuddalore); World's 2nd largest mangrove; 1100 hectares; species: Avicennia, Rhizophora

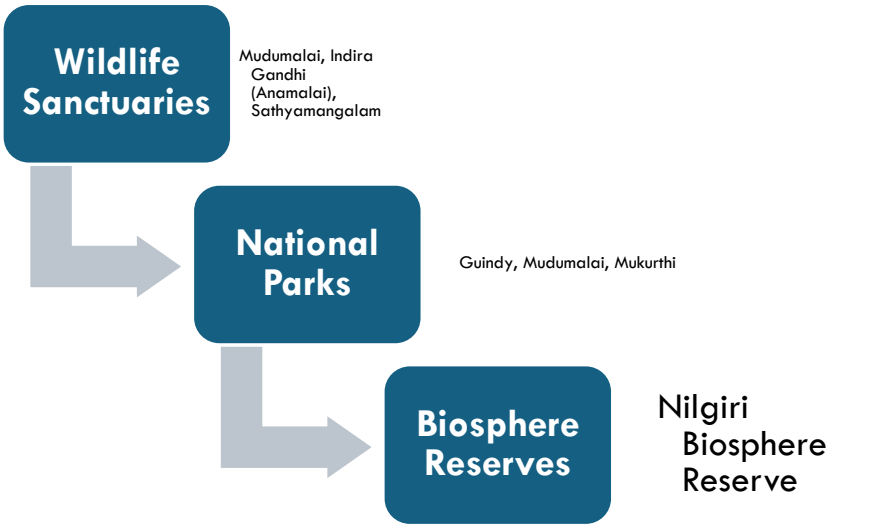
Wildlife of Tamil Nadu

Rich Fauna
Elephants, bisons, tigers, deer, monkeys, reptiles

Habitats
Mainly hill forests, reserved sanctuaries

Protection
Wildlife Sanctuaries, National Parks, and Biosphere Reserves established

Wildlife Conservation Areas of Tamil Nadu *(Quick Glance)*


Wildlife Sanctuaries
Mudumalai, Indira Gandhi (Anamalai), Sathyamangalam
National Parks
Guindy, Mudumalai, Mukurthi
Biosphere Reserves
Nilgiri Biosphere Reserve